QUESTIONS

T487IG

PART A

I Choose the best answer. **1 Mark**

1. A transistor has ……..
   a) One pn junction b) Two pn junction c) Three pn junction d) Four pn junction

2. The number of depletion layers in a transistor is …….
   a) Four b) Three c) One d) Two

3. The base of a transistor is …… doped.
   a) Heavily b) Moderately c) Lightly d) None of the above

4. In a PNP transistor, the current carriers are ……
   a) Acceptor ions b) Donor ions c) Free electrons d) Holes

5. A transistor is a …….. operated device.
   a) Current b) Voltage c) Both voltage and current d) None of the above

6. In an NPN transistor, the current carriers are …….
   a) Acceptor ions b) Donor ions c) Free electrons d) Holes

7. The emitter of a transistor is …….. doped.
   a) Lightly b) Heavily c) Moderately d) None of the above

8. In a transistor, the base current is about …….. of emitter current.
   a) 25% b) 20% c) 35% d) 5%

9. Most of the majority carriers from the emitter …….
   a) Recombine in the base b) Recombine in the emitter
   c) Pass through the base region to the collector d) None of the above.

10. In a transistor, …….
    a) $I_C = I_E + I_B$ b) $I_B = I_C + I_E$ c) $I_E = I_C - I_B$ d) $I_E = I_C + I_B$

11. A heat sink is generally used with a transistor to …….
    a) Increase the forward current b) Decrease the forward current
    c) Compensate for excessive doping d) Prevent excess temperature

12. Which of the following amplifier have lowest efficiency
    a) Class A b) Class B c) Class C d) Class AB

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